Survey on "Secure AI Architectural Rules"

Thank you for taking part in this survey on secure AI architectures!

In the following sections you will be asked to assess a set of architectural rules for securing AI systems. **Please answer based on your own experience as a security expert**.

OBJECTIVES: The outcome of this survey will be used as input in a follow-up study targeting a larger population of software practitioners.

CONTACT: If you have any questions, please contact [redacted]

Question attribute	Value
save_as_default	N

Please provide the following information about yourself

Question attribute	Value
save_as_default	N

Full name

Organization

Role

Email

How many years of experience do you have working in **Open Source Software**?

Question attribute	Value
answer_order	normal
hide_tip	1
other_position	default

< 1 year

1 to 2 years

2 to 3 years

3 to 4 years

4 to 5 years

> 5 years

How many years of experience do you have working in **Commercial (closed) Source Software**?

Question attribute	Value
answer_order	normal
hide_tip	1
other_position	default

< 1 year

1 to 2 years

2 to 3 years

3 to 4 years

4 to 5 years

> 5 years

How would you rate your **software security knowledge/skills**?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

VERY HIGH

ABOVE AVERAGE

AVERAGE

BELOW AVERAGE

VERY LOW

Based on your experience, how relevant are the following rules to the security of AI architectures?

Question attribute	Value
save_as_default	N

AI-mode switcher: An AI mode switcher can be used to let the human user control whether AI components' outcomes are followed automatically or taken as suggestions and reviewed by the user.

Multi-model decision maker: Employing a multi-model decision maker consisting of more than one different AI components can increase the system's reliability and fault-tolerance.

Redundant AI components: Deploying multiple identical AI components in parallel can ensure tolerance towards a malicious or otherwise disfunctioning AI component.

Ethical black box: AI components' outcomes and related system information should be recorded in a black box. If rewards are used as incentive for ethical behaviour, they should be recorded as well.

Continuous monitoring: AI components' outcomes should continuously be checked for their adherence to ethical requirements.

Monitoring AI components' discrepancies: If multiple redundant AI components are deployed, their outcomes should be monitored to identify discrepancies.

Adapting traditional design methods: Architectural design methods used for traditional systems (e.g., UML diagrams) can be adapted and used for AI-based systems as well and can model ethical requirements.

Access via APIs: AI-based systems can be made available via APIs instead of allowing users to run them locally to keep control over their use.

Bill of materials registry: A bill of materials registry should be integrated that keeps a machine-readable record of all used components and their properties.

Reusable components: Components should be implemented as reusable units with clearly defined boundaries and explicit interfaces. Complex dependencies and entanglement should be avoided.

Compatible data types: Incompatible data types between AI and other components should be avoided.

Single programming language: The use of multiple programming languages in the same system should be avoided where possible.

No undeclared consumers: Undeclared consumers of AI components should be avoided and access control for the outcomes of the AI components should be enforced.
Performance monitoring: There should be a component that monitors the performance of the AI components concerning their prediction accuracy.
Sufficient resources: Sufficient computing and memory resources have to be available for the AI components.
Secure data storage: When data is collected for re-training, it has to be stored securely and according to applicable regulations such as the GDPR. This includes the use of encryption techniques, a firewall between data storage and network, and enforcement of access control.
Very relevant
Relevant
Somewhat relevant
Somewhat irrelevant
Irrelevant
Not relevant at all
Thank you so much for participating in the survey! Please **CLICK SUBMIT** to record your answer.

Question attribute	Value
save_as_default	N